

Setting up the Lab Environment

7CSEF006W

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1 Lab environment

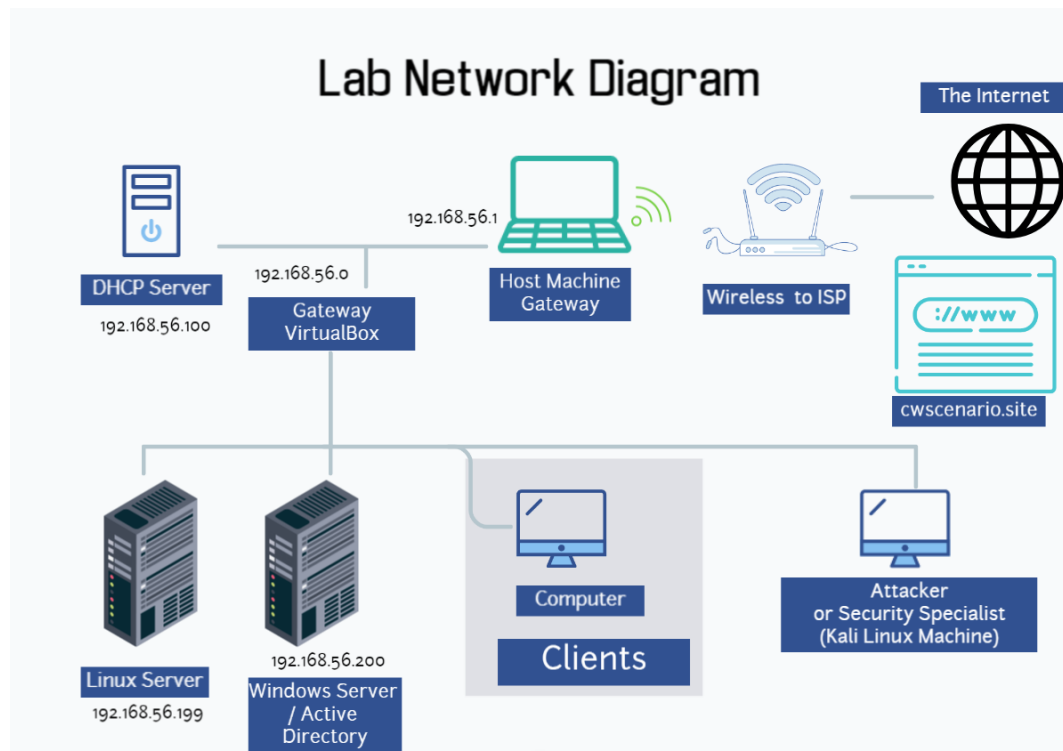


Figure 1.1: Lab environment overview

Please read this before you start

- ➡ This document explains how to download and set up the virtual machines (VMs). In most labs, the main machines are already available in `C:\VirtualMachines` ; you only need to add them to VMware.
- ➡ Before downloading a VM, first check `C:\VirtualMachines` for the relevant `.7z` archives. Most labs already provide the `.7z` files there.
- ➡ Always download from the correct local server; using the wrong server will be significantly slower.
- ➡ Only download to `C:\VirtualMachines` . If you downloaded elsewhere, move the files into that folder to avoid filling the system drive.
- ➡ Download one VM at a time. They are large files; parallel downloads can cause slowdowns or failures.
- ➡ Shut down the VM before logging out; otherwise you may not be able to use it next time.
- ➡ If a VM becomes unusable, remove it from VMware (including deleting the VM's files) and re-import it from the original archives.
- ➡ **Do not delete the `.7z` files.** They are the source for restoring or re-importing VMs quickly.

➡ During the course, several VMs are used. These can be obtained from the Virtual Machines page.

➡ Use the correct download location depending on where you are:

- ↳ On campus (Eduroam or any lab with VMware Workstation, except CLG.43 and CLG.45): <http://10.20.144.78/download/VMs/>
- ↳ In CLG.43 (Network Lab): <http://192.168.143.203/download>
- ↳ In CLG.45 (Network Lab): <http://192.168.145.206/download>
- ↳ Outside the university:
 - ↳ 1) Connect to the VPN: <https://remote2.westminster.ac.uk/>
 - ↳ 2) VPN setup guidance: <https://support.cse.westminster.ac.uk/w/index.php/VPN>
 - ↳ 3) When prompted for the portal address, use `remote2.westminster.ac.uk`
 - ↳ 4) After connecting, access: <https://download.cse.westminster.ac.uk/VirtualMachines/> and download the required VM

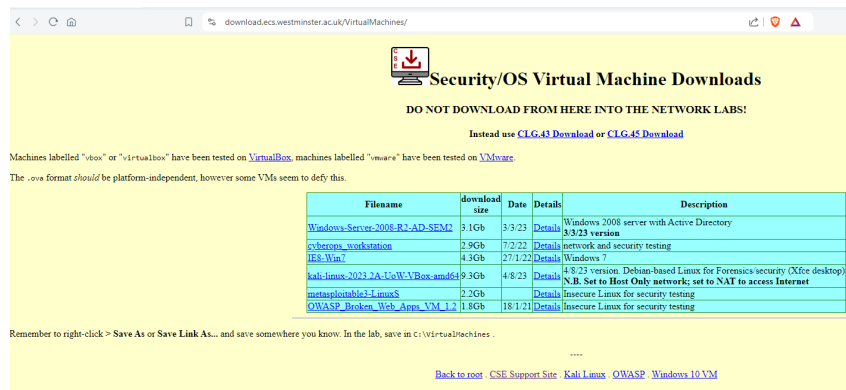


Figure 1.2: The virtual machines images download page

- ➡ Install VMware Desktop Hypervisor. On personal devices:
 - ↳ Download VMware Workstation (Windows) or VMware Fusion (macOS) from <https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>.
 - ↳ You may need to register an account.
 - ↳ As of late 2024, VMware Workstation Pro and VMware Fusion are free for personal, educational, and commercial use (no licence key required).
- ➡ On university machines, VMware Workstation is installed in CLG.43, CLG.45 and will also be available in CLG.42, LG.105, 2.110 and 2.112. Guidance: [VMware support page](#).
- ➡ Only download VMs needed for your module.
- ➡ VM set used in this module typically includes:
 - ↳ Security Analyst/Attacker: kali-linux-2025.2-vmware-amd64
 - ↳ Linux vulnerable machine: metasploitable3-LinuxS
 - ↳ Windows Server vulnerable machine: Windows-Server-2008-RD2-AD-SEM2
 - ↳ Windows Client machine: IE8-Win7

VMware: Open vs Import

- ➡ **Open an existing VM folder (.vmx + .vmdk):** In VMware Workstation, go to **File** → **Open** and select the .vmx file. The VM is added directly to your library.
- ➡ **Import a packaged VM (.ova):** Go to **File** → **Open** and select the .ova. VMware will import and create the required .vmx and .vmdk files.
- ➡ Most Cyber Security VMs are provided as ready-built VMware folders (.vmx + .vmdk); importing .ova is rarely needed.

- ➡ After downloading, install VMware Workstation/Fusion on your device if needed.

- ➡ Add or import VMs from `C:\VirtualMachines` (or from your download location).
- ➡ Start VMs from the VMware interface using the **Start** button or by double-clicking the VM in the library.

Keyboard layout note

- ➡ On Kali, if using a British English keyboard, change the keyboard layout (default is US), to avoid confusion locating certain characters.

1.1 Virtual machines setup

- ➡ Kali Linux credentials:
 - ↳ username: `kali`
 - ↳ password: `kali`
- ➡ Metasploitable3 (Linux) credentials:
 - ↳ username: `linuxs`
 - ↳ password: `LinuxS`
- ➡ Windows AD server login details will be provided when required.

2 Network Setup

2.1 Virtual Network Editor - on your own machine

Network Configuration Note

- ➡  Most labs require the **isolated host-only network** unless the instructions specify otherwise.
- ➡ Creating or modifying a **virtual network** using the VMware **Virtual Network Editor** requires **administrator access** and can only be done on your **own personal machine**.
- ➡ On university lab machines, these host-only networks are already pre-configured - do **not** attempt to create or modify them.

- ➡ If you are using your own machine, follow these steps to **create a host-only virtual network**:

↳ Open **VMware Workstation** → **Edit** → **Virtual Network Editor** (Fig.2.1).

- ↳ Click **Change Settings** (admin rights required) (Fig.2.5).
- ↳ Add a **Host-only** network (e.g., VMnet1) (Fig.2.1).
- ↳ Enable DHCP (**Use local DHCP to distribute IP addresses**).
- ↳ Optional: For consistency with university lab setups, open **DHCP Settings** and set the starting IP to 192.168.144.100 (Fig.2.3).

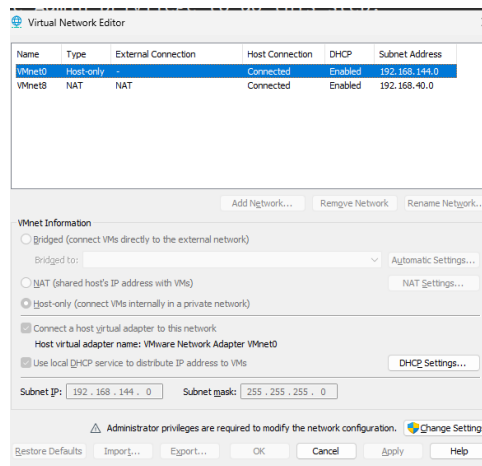


Figure 2.1: Virtual Network Editor: **Edit** → **Virtual Network Editor**

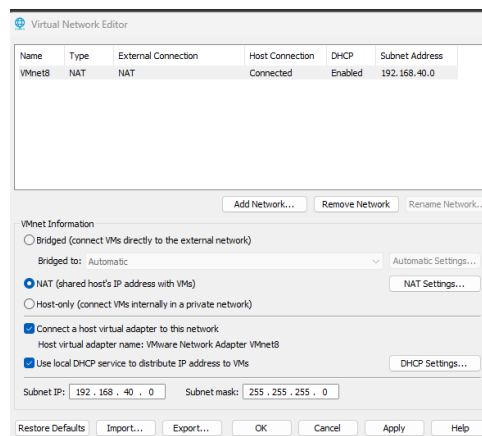


Figure 2.2: Add Network

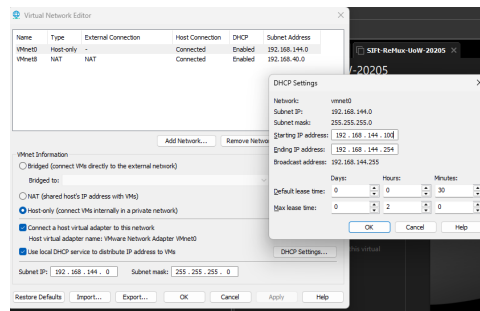


Figure 2.3: Enable DHCP

2.2 Host-only Connected VM – Isolated Network

➡ After the host-only network exists (on your machine or on the pre-configured lab machines), you must configure each VM to use it:

- ↳ Select the VM, then go to **VM → Settings**.
- ↳ Shortcut: Press **Ctrl+D**, or right-click the VM and choose **Settings**.
- ↳ Under **Network Adapter**, select **Host-only**.

⚠ This step does **not** require administrator privileges and must be done by **everyone**, including students using university lab PCs.

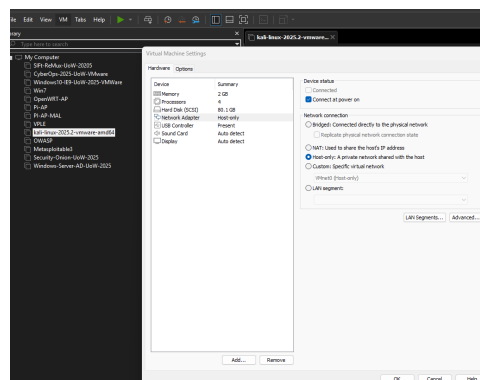


Figure 2.4: Host-only Adapter Settings

➡ To check your IP address on Linux (Kali or Metasploitable3):

```
➤ ip a
```

↳ Legacy alternative:

```
➤ ifconfig
```

➡ To check your IP address on Windows: open **cmd** and run:

```
➤ ipconfig
```

2.3 Internet Connected VM

Internet Connected Network

- ➡ Sometimes only the Kali VM needs Internet access.
- ➡  Do not connect the Windows 7, windows AD or the metasploitable VMs to the Internet.

- ➡ To temporarily switch a VM's network to access the Internet via the host:
 - ↳ Select the VM → **VM** → **Settings** → **Network Adapter** (Fig.2.5).
 - ↳ **Select NAT** to allow the VM to access the Internet through the host's connection.
 - ↳ **Select Host-only** to switch the VM back to the isolated lab network.
 - ↳ This adapter change does **not** require administrator privileges and can be done on both personal machines and university lab PCs.

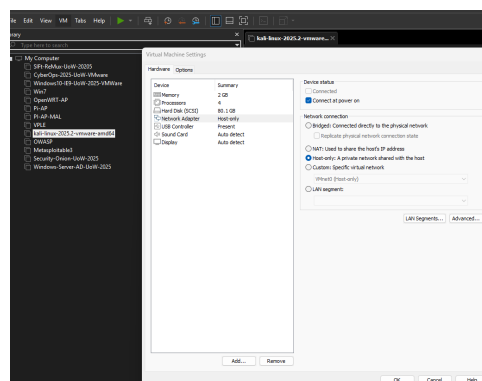


Figure 2.5: Choose Settings

3 Test connectivity

- ➡ To test connectivity from Kali to the vulnerable machine using ICMP:

```
 ping 192.168.144.100
```
- ➡ A host-only network isolates your VMs from the Internet while allowing them to communicate with each other.
- ➡ For any two devices to communicate, they must be on the same IP network (same subnet).
- ➡ To test an HTTP service from Kali, open a browser and navigate to the vulnerable machine's IP:

http://192.168.144.100

Resetting Kali's Network After Switching NAT $\leftrightarrow$ Host-only

➡ When switching the VM's network adapter between **NAT** and **Host-only**, the IP address may not update automatically. Reset the interface manually to obtain a fresh IP address:

```
>- sudo ip link set eth0 down
```

```
>- sudo ip link set eth0 up
```

```
>- sudo dhclient eth0
```

➡ Finally, confirm the assigned IP address:

```
>- ip a
```

4 Setting up Python in Linux

➡ To install Python 3 and common tools on Debian-based systems (e.g., Kali/Ubuntu):

```
>- sudo apt-get update
```

```
>- sudo apt-get install -y python3 python3-venv python3-pip
```

➡ To remove unused packages afterwards (optional):

```
>- sudo apt-get autoremove
```

Important note on NAT connection when working from home - resolvconf

➡ Some activities require only the Kali VM to be online (NAT).

➡ The university Kali image is set to work from within the university labs and avoid blocked kali domain to update the repository.

➡ For this reason, you need to switch the dns resolver to home network.

➡ A script `uow-kali/scripts/switch-dns.sh` is available to make switching resolver profiles easier.

➡ If not already present, it can be downloaded from github.com/azelhajjar/uow-kali or cloned directly:

```
>- git clone https://github.com/azelhajjar/uow-kali
```

➡ Usage examples:

```
>- sudo ./switch-dns.sh uni # switch to University DNS
```

```
>- sudo ./switch-dns.sh home # switch to Home DNS
```

```
>- sudo ./switch-dns.sh status # check active profile
```


5 Problems that you can possibly face

- ➡ Ensure your VM's virtual network is actually connected (link present in VMware).
- ➡ On Kali, network interface changes sometimes do not apply until NetworkManager is restarted:

```
❏ sudo service network-manager restart
```